COURSE OUTLINE

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Troubleshooting Automation with Claude

From ServiceNow acknowledgement to reason, action and next action

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Network operations, load balancer and data centre engineers, NOC and service desk teams, IT automation leads in India and worldwide.

Prerequisites

- Basic networking and load balancer concepts (VIP, pool, HA)
- Comfort with a CLI such as F5 tmsh or NetScaler shell
- Basic Python and Git
- Familiarity with ServiceNow incidents is helpful, not required

What you will be able to do

- Explain the incident-to-resolution workflow in ServiceNow and where an agent can and cannot act
- Set up Claude Code, the Claude Agent SDK and MCP servers scoped to incident triage with read-only tools by default
- Pull incidents through the ServiceNow Table API, draft the acknowledgement mail and ask for the primary troubleshooting details that are missing
- Collect typed evidence from F5, NetScaler, Radware, other load balancers (Anaya LB) and Cisco ACI and have Claude suggest ranked CLI troubleshooting steps as structured JSON
- Enforce read-only versus state-changing command classes, human approval and a full audit trail into ServiceNow work notes
- Generate and post the reason, action and next action template and ship a tested troubleshooting page

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Tools and technologies

Claude Code | Claude Agent SDK and MCP | ServiceNow Table API and Flow Designer | Python | Pydantic | FastAPI | F5 tmsh and iControl REST | Citrix NetScaler NITRO | Radware Alteon | Cisco APIC REST API | pytest | GitHub Actions

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DAY 1 3 modules

MODULE 1

The troubleshooting workflow and the ServiceNow incident lifecycle

- Incident states, assignment groups, acknowledgement mail and the first troubleshooting reply as they happen today
- Mapping the manual workflow: ticket lands, details requested, evidence gathered, steps suggested, resolution note written
- Where an agent may act (read, draft, explain) and where it may not (change a device, close a ticket) without a person
- The lab: a ServiceNow developer instance, lab load balancers and a Cisco ACI sandbox

HANDS-ON LAB
Walk one real-style incident from creation to resolution by hand and record every decision an engineer makes.

DELIVERABLE
Annotated workflow map and automation boundaries.

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MODULE 2

Setting up Claude Code, the Agent SDK and MCP for incident triage

- Claude Code project configuration, CLAUDE.md conventions and permission rules for this workflow
- Claude Agent SDK: sessions, tool definitions, structured JSON outputs and retries
- MCP servers for ServiceNow and for device read-only queries, with tool allowlists
- Keeping credentials in environment and vault, never in prompts or transcripts

HANDS-ON LAB
Build and register a read-only ServiceNow MCP server and a read-only device MCP server and call them from an Agent SDK script.

DELIVERABLE
Working agent environment with scoped tools.

MODULE 3

Incident intake through the Table API and the acknowledgement mail

- Querying incident and sys_journal_field through the ServiceNow Table API and validating records with Pydantic
- A primary troubleshooting checklist: affected VIP or service, symptom, start time, scope, last change, error text
- Claude compares the ticket against the checklist and lists exactly which details are missing
- Drafting the acknowledgement mail with the missing-details questions; a person reviews and sends

HANDS-ON LAB
Ingest five synthetic incidents, generate the missing-details list and acknowledgement draft for each, and post it as a work note.

DELIVERABLE
Intake service and acknowledgement mail generator.

DAY 2 4 modules

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MODULE 4

Evidence collection from load balancers and ACI

- F5: tmsh show ltm virtual and pool, iControl REST /mgmt/tm/ltm and /mgmt/tm/sys for stats and logs
- Citrix NetScaler NITRO API: lbvserver, servicegroup and HA node status; Radware Alteon and other load balancers (Anaya LB) through their CLI
- Cisco APIC REST API: fault, health score and endpoint lookups for the affected tenant
- Typed evidence records with device, timestamp, command and raw output; nothing inferred without data

HANDS-ON LAB
Write deterministic collectors for each platform and run them against the lab devices for one incident.

DELIVERABLE
Evidence bundle per incident.

MODULE 5

Claude suggests troubleshooting steps as structured JSON

- Prompt design: ticket plus evidence in, ranked likely causes out, each with a confirming CLI command and expected result
- JSON schema for a suggested step: platform, command, class (read_only or state_changing), rationale, evidence citations
- Testing the prompt with missing, stale and contradictory evidence; unknowns must be marked
- Comparing suggestions against a vendor-documented playbook for the same symptom

HANDS-ON LAB
Build the suggestion agent, run it on three incidents and score its first-step accuracy against the engineer's manual notes.

DELIVERABLE
Troubleshooting step suggestion agent.

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MODULE 6

Guardrails, approvals and audit for incident automation

- Command classification and allowlists: read-only commands may run in the lab, state-changing commands always wait
- Approval record: who approved, which command, which device, when, and the output captured
- Writing every action to an audit log and to ServiceNow work notes through the Table API
- Failure modes: wrong device, wrong incident, hallucinated command; detection and fail-closed behaviour

HANDS-ON LAB
Add the approval gate and audit writer, then try to push a state-changing command through without approval and watch it refuse.

DELIVERABLE
Approval gate and audit trail.

MODULE 7

Running approved steps and reading the result

- Executing approved commands through SSH and REST wrappers with timeouts and output capture
- Claude interprets command output against the expected result and proposes the next step
- Loop control: maximum steps, stop conditions and escalation to a senior engineer
- ServiceNow Flow Designer trigger that starts the loop when an incident is assigned to the group

HANDS-ON LAB
Run the suggest, approve, execute, interpret loop end to end on one incident until the cause is confirmed.

DELIVERABLE
Supervised troubleshooting loop.

DAY 3 3 modules

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MODULE 8

The reason, action and next action template

- Template fields: reason (root cause with evidence), action (what was done, by whom, approved by whom), next action (owner and due date)
- Claude drafts the template from the evidence bundle and command outputs with a citation for every claim
- Posting resolution notes and close codes to the incident and creating a knowledge article draft
- Template variants per symptom family: VIP down, pool member flapping, HA sync failure, certificate expiry

HANDS-ON LAB
Generate the template for the resolved incident, review it, post it to ServiceNow and create the knowledge article.

DELIVERABLE
Reason, action and next action generator.

MODULE 9

The troubleshooting page: service, tests and delivery

- FastAPI troubleshooting page: incident queue, evidence view, suggested steps, approve and reject buttons, audit tab
- pytest suites for intake, classification and template generation with recorded device output as fixtures
- GitHub pull requests and Actions: lint, tests and a prompt regression check on every change
- Operating the page: runbook, on-call handover and what to do when the agent is wrong

HANDS-ON LAB
Assemble the page, write the regression suite and merge through a reviewed pull request with Actions green.

DELIVERABLE
Tested troubleshooting page and runbook.

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MODULE 10

Capstone: incident to resolution note, end to end

- A new incident lands in ServiceNow; the team runs the full workflow from acknowledgement to posted template
- Acceptance criteria review: no device change without approval, every claim cited, every action logged
- Demo and limitations statement

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end troubleshooting automation and runbook.

Capstone

Capstone project

A repeatable troubleshooting workflow that picks up a ServiceNow incident, asks for the missing primary troubleshooting details, collects evidence from F5, NetScaler, Radware, other load balancers (Anaya LB) and Cisco ACI, suggests ranked CLI steps, runs only approved commands, and posts a reason, action and next action template with a full audit trail.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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